Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=3;i++)
4      (i-1)[arr] = (i%3);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 1;
8   }
9  int main(){
10   int arr[4] = {0,0,0,0};
11   func1(arr+1);
12   for(int i=0; i<4; i++)
13          printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<4; i++)
17          printf("%d ", i[arr] );
18   return 0;
19 }
```

i=1 1
2 2
3 0.

arr[i-1]

1 20.

0120.

arr[i]

1120.

Output
0120
1120

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 1;
 a = &temp;
}
```

Explain your answer

Using above definition of func2, output of line 16-17 will be 0120. when we write a = &temp, a points to temp instead of arr[0] and hence arr remains unaltered.

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

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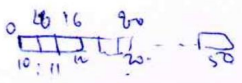
$a_1 - a_2$ gives difference in address of arr2[50] & arr2[10]. Address of consecutive elements of arr2 differ by 8 bytes (due to long) $a_1 - a_2$ is difference b/w 40 elements of array. Similarly, $b_1 - b_2$ also produces diff of 40 elements but since it is of type int, size 4.



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